

A Literature-Implied Answer to the Kriete–MacCluer Beurling–Carleson Regime

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Status. This is a lightweight literature-implied status note:

`literature_implied_answer.`

It records that the future-work direction flagged in Malman, *Revisiting mean-square approximation by polynomials in the unit disk*, arXiv:2304.01400, is answered in the principal standard-weighted Bergman setting, and in the explicit subexponential gauge model, by Malman’s later paper *Weighted Korenblum–Roberts theory*, arXiv:2503.20054.

Source question. In arXiv:2304.01400, Introduction, subsection “Work of Kriete and MacCluer”, the source explains that Kriete and MacCluer’s Conjecture 2 is settled by the paper in the fast-decay regime

$$\liminf_{x \rightarrow 0^+} x \log(1/G(x)) > 0.$$

The next paragraph records their Conjecture 1 for the limiting-zero regime. The expected rule is analogous to the fast-decay theorem, but with intervals replaced by Beurling–Carleson sets, and the source says that a conclusive answer is future work.

Supporting theorem. In arXiv:2503.20054, the supporting paper defines the associated Beurling–Carleson family

$$\text{Assoc}(w) = \left\{ E \in \text{BC} : \int_E \log w \, dm > -\infty \right\},$$

then defines the core of w as the maximal-measure union generated by such sets and the residual part as the carrier of w outside the core. Its Theorem T:ThomsonDecompTheorem states that, for

$$d\mu = dA_\alpha + w \, dm, \quad \alpha > -1,$$

and every $0 < t < \infty$,

$$\mathcal{P}^t(\mu) = \mathcal{P}^t(dA_\alpha + w_c \, dm) \oplus L^t(w_r \, dm),$$

where w_c and w_r are the core and residual restrictions of w , and the first summand is irreducible. Thus full splitting off the boundary Lebesgue part occurs exactly when the Beurling–Carleson core contribution vanishes, and in general the theorem gives the promised Beurling–Carleson replacement for the interval residual decomposition from the fast-decay regime.

Gauge version and match to the 2023 discussion. The same supporting paper also formulates the generalized measure

$$d\mu = G_{ah} dA + w dm, \quad G_{ah}(z) = \exp\left(-a \frac{h(1-|z|)}{1-|z|}\right),$$

with the associated h -Beurling–Carleson family. It states explicitly that for $h(x) = x^\beta$, $0 < \beta < 1$, one has

$$\mathcal{P}^t(\mu) = \mathcal{P}^t(G_{ah} dA + w_c dm) \oplus L^t(w_r dm),$$

again with an irreducible first summand. This includes

$$G_{ah}(z) = \exp\left(-\frac{a}{(1-|z|)^{1-\beta}}\right),$$

which is the subexponential model discussed in the sharpness section of arXiv:2304.01400.

Why this is literature-implied rather than already-answered. The supporting paper cites arXiv:2304.01400 and explains the relation between the fast interval regime and the slower associated-Beurling–Carleson regime, but it does not label its theorem as “Kriete–MacCluer Conjecture 1”. The identification is therefore a direct reformulation by this run: matching the source’s requested replacement of intervals by Beurling–Carleson sets with the later core/residual decomposition theorem.

Scope limitations. This packet should not be used as a claim that every radial disk weight G with

$$\lim_{x \rightarrow 0^+} x \log(1/G(x)) = 0$$

is covered. The supporting theorem covers the standard weighted Bergman area case and the stated gauge families, including $h(x) = x^\beta$. Any arbitrary radial-weight extension would still require checking the regularity/gauge hypotheses or a separate theorem.

References.

- B. Malman, *Revisiting mean-square approximation by polynomials in the unit disk*, arXiv:2304.01400.
- B. Malman, *Weighted Korenblum–Roberts theory*, arXiv:2503.20054.
- T. L. Kriete and B. D. MacCluer, *Mean-square approximation by polynomials on the unit disk*, Transactions of the American Mathematical Society 322 (1990), 1–34.